

Project Olympus · Unit 1: The Language of Change

Student practice sheets. Paper first. · The Language of Everything · David Balyeat, with Claude

Week 1 · Day 1: The Machine

Functions: something goes in, a rule happens, something comes out

Warm-up

A server earns a **\$16.50/hour** base. Four hours of work: base pay?

Practice

1. $f(x) = 3x + 2$. Find **$f(5)$** .

2. $g(x) = x^2 - 1$. Find **$g(4)$** .

3. Server pay: $P(s) = 82.5 + 0.18s$. A slow night, \$400 in sales. **$P(400)$** = ?

4. Fortnite phase 3 storm: 5 HP/s. $H(t) = 100 - 5t$. Your health after 12 seconds in the storm: **$H(12) = ?$**

5. Spider-Man's web cartridge holds **200 shots**. Each swing uses **3**. Which rule gives shots left after n swings?

- A. $S(n) = 200 + 3n$
 - B. $S(n) = 200 - 3n$
 - C. $S(n) = 3n - 200$
 - D. $S(n) = 200 \div 3n$
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6. Using $S(n) = 200 - 3n$, how many shots are left after **50 swings**?

7. In Minecraft, walking speed is **4.317 blocks per second** (a block is one meter). Distance walked: $D(t) = 4.317t$. **$D(60) = ?$** (Nearest whole block.)

8. In Harry Potter's world, 29 Knuts make a Sickle and 17 Sickles make a Galleon, so a Galleon is $17 \times 29 = 493$ Knuts. Knuts in g Galleons: $K(g) = 493g$. **$K(3) = ?$**

Week 1 · Day 2: Pictures of Relationships

Descartes, the fly on the ceiling, and why a graph is worth a thousand numbers

Warm-up

In your head: the point **(3, -2)** is where, relative to the axes?

- A. Right of the y-axis, above the x-axis
- B. Right of the y-axis, below the x-axis
- C. Left of the y-axis, below the x-axis
- D. Left of the y-axis, above the x-axis

Practice

1. In the point **(4, 9)**, what is the x-coordinate?

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2. $f(x) = 2x + 1$. Which point is on its graph?

- A. (3, 7)
 - B. (3, 6)
 - C. (7, 3)
 - D. (2, 4)
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3. Messi at 8.5 m/s: how far at **t = 4**?

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4. From the graph of $d = 8.5t$: at what time is Messi at **25.5 m**?

5. Yamal at **9.86 m/s** (35.5 km/h, fastest of the Spain squad at the 2026 World Cup): how far after 3 seconds? One decimal.

6. Two distance-time lines on the same axes: Yamal's and Messi's. Which is steeper, and why?

- A. Yamal's, because he covers more meters per second
 - B. Messi's, because he has more goals
 - C. They're the same; both are straight
 - D. Neither; steepness depends on the units
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7. Every point that sits *on the y-axis* has what x-coordinate?

Week 1 · Day 3: Slope Is a Rate

Rise over run, and why it is secretly velocity

Warm-up

Convert **36 km/h** to meters per second.

Practice

1. Slope of the line through **(1, 8.5)** and **(3, 25.5)**.

2. Storm phase 3: health goes from **(0 s, 100 HP)** to **(20 s, 0 HP)**. Slope?

3. Convert Ronaldo's **33.95 km/h** to m/s. Two decimals.

4. Storm phase 2 shrinks for 75 seconds. Say the radius goes from **1,200 m to 600 m**. Slope, in m/s?

5. A double-leg shot: **1.5 m in 0.5 s**. Average speed?

6. Minecraft sprint-jumping averages **7.127 m/s**. The line through (0, 0) and (10, ?): what's the missing y? Two decimals.

7. Line A has slope 4. Line B has slope 9. Both are distance vs. time. Which is faster?

- A. A
 - B. B
 - C. Same speed
 - D. Can't tell without the intercept
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8. A distance-time line with **zero slope** (flat) means:

- A. Standing still
 - B. Moving backward
 - C. Infinite speed
 - D. Moving at 1 m/s
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Week 1 · Day 4 · Lab Day: Lab Day: Galileo's Ramp

Measure motion with your own hands, plot it, and find the slope

Warm-up

A ball rolls **2 meters in 4 seconds**. Average speed?

Practice

1. Using the sample data: average speed over the whole run (1.6 m in 2.0 s)?

2. Slope between the **last two** sample points, (1.5, 0.9) and (2.0, 1.6)?

3. Slope between the **first two** sample points, (0, 0) and (0.5, 0.1)?

4. The slope keeps increasing along the run. The ball is:

- A. Speeding up
 - B. Slowing down
 - C. Moving at constant speed
 - D. Stopped
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5. Times are in A2:A6 and distances in B2:B6. Which formula gives the slope?

- A. =SLOPE(A2:A6, B2:B6)
 - B. =SLOPE(B2:B6, A2:A6)
 - C. =SLOPE(A2:B6)
 - D. =RISE(B2:B6)/RUN(A2:A6)
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Week 2 · Day 1: $y = mx + b$

Every straight line has a starting point and a rate. Learn to read both.

Warm-up

$P = 82.5 + 0.18s$. If sales are **zero**, what's the pay?

Practice

1. $y = 4x + 7$. The slope is:

2. $y = 4x + 7$. The y-intercept is:

3. A line passes through **(0, 12)** with slope **3**. Its equation?

A. $y = 12x + 3$

B. $y = 3x + 12$

C. $y = 3x - 12$

D. $y = x + 15$

4. A 6-hour shift at \$16.50/hour gives \$99 base, plus 18% of sales. Pay as a function of sales s :

- A. $P = 99s + 0.18$
 - B. $P = 0.18s + 99$
 - C. $P = 18s + 99$
 - D. $P = 0.18(s + 99)$
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5. Filament: $C = 20k + 5$ (dollars, k in kg). Cost of **3.5 kg**?

6. Rubik's average: $T = 60 - 2w$. After **10 weeks**?

7. Storm: $r = 1500 - 6.67t$. Radius after **60 seconds**? Nearest meter.

8. Which line is **decreasing** (goes down as x increases)?

A. $y = 2x + 5$

B. $y = -2x + 5$

C. $y = 5$

D. $y = x$

Week 2 · Day 2: Solving for x

An equation is a balance. Undo what was done. And the trick, explained twice.

Warm-up

$3x = 21$. What is x ?

Practice

1. $4x - 6 = 18$

2. $\frac{x}{5} + 3 = 10$

3. $2x + 7 = 4x - 5$

4. Storm phase 4 (7 HP/s): $H = 100 - 7t$. Solve for the time when $H = 30$. One decimal.

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5. Server: $P = 0.18s + 99$. How much in sales to take home **\$200**? Nearest dollar.

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6. Rubik's: $T = 60 - 2w$. When does the average hit **30 s**?

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7. Design check: "Multiply your number by 4, add 12, divide by 4, subtract your number."
Everyone gets:

.....

8. Diophantus's tombstone (paraphrased): "He was a boy for one-sixth of his life. After one-twelfth more, he grew a beard. After one-seventh more, he married. Five years later a son was born. The son lived half as long as his father. Four years after the son died, Diophantus died." How old was Diophantus?

Week 2 · Day 3: Equations in the Wild

Turning sentences into equations, from Minecraft to Gringotts

Warm-up

Knuts per Galleon: $17 \times 29 = ?$

Practice

1. Minecraft walking speed is **4.317 m/s**. How long to walk **1,000 blocks** to your friend's base? Nearest second.

2. Sprinting is **5.612 m/s**. How many seconds does sprinting save on those 1,000 blocks? Nearest second.

3. Web cartridge: $S(n) = 200 - 3n$. After how many swings are **50 shots** left?

4. Convert 2 Galleons, 5 Sickles, 20 Knuts entirely into Knuts.

5. In *Percy Jackson*, the Gray Sisters' taxi charges (let's say) a 1-drachma flat fee plus 2 drachmas per mile. A ride costs 15 drachmas. How many miles?

6. Hudson wants \$1,000 for a resin printer. Each restaurant shift nets about \$150. How many shifts? (Whole shifts; you can't work 0.67 of one.)

7. Your Rubik's average is 45 s, improving 1.5 s per week. Weeks until you average 30 s?

8. In *Zelda*, say Link runs about **6 m/s** on flat ground. A shrine is **300 m** away. Time to get there?

Week 2 · Day 4 · Lab Day: Lab Day: The Tip Calculator

Two real Sheets models, one chart, and Beat Dad week 2

Warm-up

18% of \$250?

Practice

1. With hours = 5 and base rate = 16.50, what does the Base pay cell ($=B1*B2$) show?

2. Sales = 700, tip rate = 18%. The Tips cell shows:

3. In the sprint sheet, seconds for Yamal (9.86 m/s) to cover **30 m**: $=30/9.86$ shows (2 decimals):

4. Same for Messi (8.58 m/s), 30 m, 2 decimals:

5. On the sprint chart at $t = 2$ s, whose line is on top?

- A. Messi
 - B. Yamal
 - C. They're equal
 - D. Depends on the axes
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Week 3 · Day 1: Two Lines Meet

Systems of equations, Achilles, the tortoise, and whether Yamal catches Ronaldo

Warm-up

Is the point **(2, 5)** on *both* lines $y = 2x + 1$ and $y = x + 3$?

- A. Yes, on both
- B. Only on the first
- C. Only on the second
- D. Neither

Practice

1. Achilles (10 m/s) vs. tortoise (0.1 m/s, 100 m head start). Solve $10t = 0.1t + 100$. Two decimals.

2. The chase: $9.86t = 9.43t + 5$. Solve for t (two decimals).

3. How far has Yamal run when he catches Ronaldo? (Use $t = 11.63$.) Nearest meter.

4. The pitch is 105 m long. With a 5 m head start, does Yamal catch Ronaldo before the end line?

- A. Yes, at about 60 m
 - B. No; the catch point is past the end line
 - C. Yes, exactly at the line
 - D. Can't tell from the model
-

5. $y = 2x + 1$ and $y = -x + 7$. The x-coordinate of the intersection?

6. ...and the y-coordinate?

7. Two lines with the **same slope** and **different intercepts** (parallel). How many solutions does that system have?

- A. One
 - B. Two
 - C. None
 - D. Infinitely many
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Week 3 · Day 2: Substitution and Elimination

Two ways to solve a system exactly, and the Lego mix problem

Warm-up

If $y = 3x$ and $x + y = 12$, what is x ?

Practice

1. $y = 2x + 1$ and $x + y = 10$. Find x .

2. ...and y .

3. $2x + y = 11$ and $x - y = 1$. Find x by elimination.

4. Lego mix: 100 pieces, bricks at 10¢ and plates at 3¢, total \$7.20. How many **bricks**?

5. Fortnite: you have **500 building materials** total, wood and brick. You have **100 more wood than brick**. How much **brick**?

6. Scoring: takedowns 2 points, mounts 4 points. **5 scores, 14 points**. How many **mounts**?

7. Camp Half-Blood store: 3 ambrosia + 2 nectar = 13 drachmas. 1 ambrosia + 2 nectar = 7 drachmas. Price of **one ambrosia**?

Week 3 · Day 3: The Whole Language

Review, the Feynman technique, and the mistakes everyone makes

Warm-up

Slope of $y = -3x + 8$?

Practice

1. $f(x) = -2x + 9$. Find $f(3)$.

2. Slope through **(2, 3)** and **(6, 15)**.

3. Slope 3, passes through $(0, -4)$. Equation?

A. $y = -4x + 3$

B. $y = 3x - 4$

C. $y = 3x + 4$

D. $y = -3x - 4$

4. $6x - 9 = 3x + 12$

5. Mbappé, **37.6 km/h**, fastest at the 2026 World Cup. In m/s, two decimals:

6. Storm phase 4: 7 HP/s. From 100 HP, how many seconds until zero? One decimal.

7. $y = x + 4$ and $y = 3x$. The x at the intersection?

8. 1,000 Knuts is how many **whole Galleons**? (493 Knuts per Galleon.)

9. Minecraft sprinting (**5.612 m/s**): time for 500 blocks, one decimal.

10. A shot covers **1.8 m in 0.6 s**. Speed?

Week 3 · Day 4 · Lab Day: Lab Day: The Catch

Labor 1, Quest 2, the family show, and the Cartographer badge

Warm-up

15% of 60?

Practice

1. Warm the engine for the Labor: convert **35.5 km/h** (Yamal) to m/s, two decimals.

2. Yamal's speed minus Ronaldo's ($9.86 - 9.43$): the **closing speed**, in m/s.

3. Using $\text{gap} \div \text{closing speed}$: with a **10 m** gap, how long to close it? One decimal.

Labor 1: The Catch

Cristiano Ronaldo (9.43 m/s, his 2018 World Cup number) has a **6-meter head start** on Lamine Yamal (35.5 km/h at the 2026 World Cup). Both at top speed, straight line, full pitch. Does Yamal catch him? Prove it three ways: algebra here, a graph on paper, a table in Sheets. No hints.

Part 1. Part 1. Convert Yamal's 35.5 km/h to m/s (two decimals).

Part 2. Part 2. Write both distance equations (Ronaldo: $d = 9.43t + 6$; Yamal: $d = 9.86t$), set them equal, and solve for the catch time t . Two decimals.

Part 3. Part 3. How far from Yamal's starting point is the catch? Nearest meter.

Part 4. Part 4. The pitch is 105 m long. Verdict?

- A. Yamal catches him around midfield
 - B. Yamal catches him just before the end line
 - C. Yamal does not catch him; Ronaldo reaches the end line first
 - D. They arrive together
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Part 5. Part 5. What head start (in meters, one decimal) would make Yamal catch Ronaldo *exactly* at the 105 m end line? (Hint you're allowed: how long does Yamal take to run 105 m? How far does Ronaldo get in that time?)
